

10.2.2 Kjeldahl and Dumas Method Comparison

The experimental design is classical, formed by $I = 3$ series with $J = 3$ replicates per series. The number of validation materials is $K = 7$. These are commercial dairy products such as yoghurt, fermented milk, or fresh cheese, except for the most concentrated material which is gelatin. These are, therefore, matrices without standard additions, which explains the absence of levels between 10 and 30 g of nitrogen/100 g, as dairy products do not naturally reach such concentrations.

The same experimental designs were applied for both Kjeldahl and Dumas methods. Since the standardized Kjeldahl method is considered the reference method capable of defining the reference value, the target content assigned to each material was established by the average of $I \times J = 9$ replicate measurement values performed with the Kjeldahl method.

The acceptance interval is $\pm 5\%$. This is narrow because manufacturers require the results of the two methods to be remarkably close for regulatory labeling reasons. To be converted into protein, the Kjeldahl nitrogen content is multiplied by 6.25: a simple difference of 0.16 g of Kjeldahl nitrogen becomes 1 g of total protein. It can create problems in the case of verification of the claimed nutrient composition.

According to the MAP of the Dumas method in Figure 10.5, a systematic overestimation of the concentration is observable when compared to the Kjeldahl reference method is observed. In absolute value, this bias varies between 0.14 and 0.25 g/100 g depending on the matrix type.

With available data, it is possible to build the MAP of the two methods and obtain their respective uncertainty functions. They are combined in Figure 10.6. They have similar shapes, the Kjeldahl method having even slightly lower performance when compared to Dumas's method. At low concentrations, the bias is between 5 and 7%, and the relative uncertainty is of the same magnitude.

Therefore, the bias becomes very bothersome when controlling low concentration foods, such as milk or yoghurt. Because the Kjeldahl method is time-consuming and polluting, a trend has been to reduce the time of mineralization in laboratories. In addition, various toxic or polluting catalysts used in the past have been replaced by others that do not have the same efficiency.

After this study, the Kjeldahl method has been suspected of not being as accurate as its *gold standard* label would suggest. The various materials used to construct the profile of the Dumas method were resampled and reanalyzed with the Kjeldahl operating procedure, but extending the duration of mineralization.

The new profile obtained with these modified reference measures is presented in Figure 10.7. The systematic bias has almost disappeared, and the method meets the requirements of the end users over the whole validation range. In this case, it is not mandatory to add to the MU the uncertainty of a correction factor: new reference values were assigned to the validation materials, but Dumas' measurements were not modified.

This example demonstrates that, when validating an alternative method against a reference method, the existence of a possible bias induced by the so-called reference method is always possible and must be carefully evaluated. This is misleading when

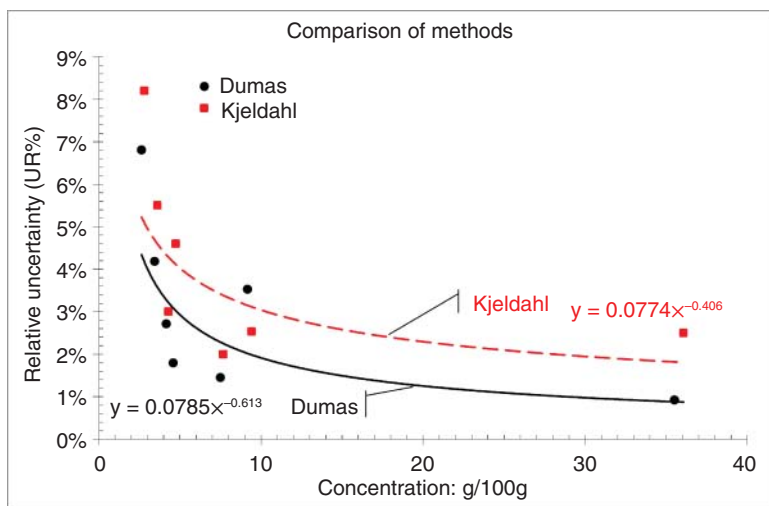


Figure 10.6 Comparison of the uncertainty functions of both methods used to determine total nitrogen in foods.

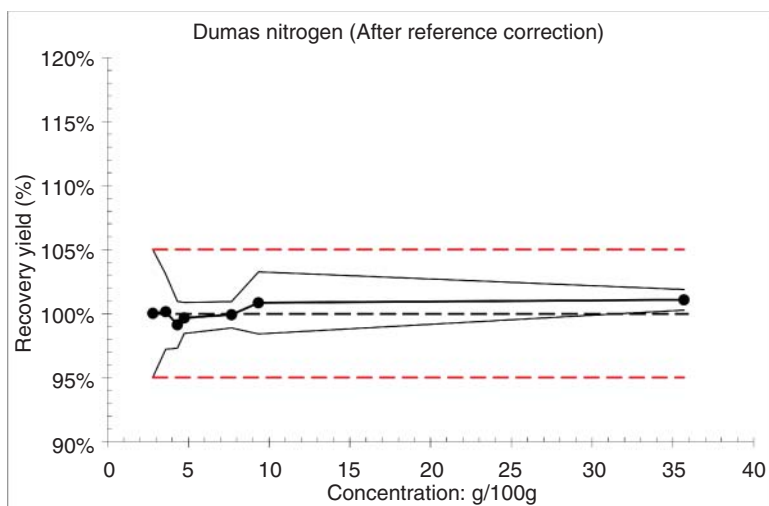


Figure 10.7 Accuracy profile of Dumas's method applied to dairy products compared to improved Kjeldahl method.

it is a *per se* defined method. This drawback only appears when the analyte is defined by the method. To detect this additive bias, it was necessary to focus on low concentrations because the trueness was expressed as a recovery rate.

Today the Dumas method is considered accurate and represents a credible alternative to the Kjeldahl method for analyzing the protein content of many foods. Even international standards address this method, such as the standard ISO 16634-2:2016 for cereals, pulses, and milled cereal products.

To conclude on the alternative Kjeldahl methods, there are other operating procedures based on near-infrared spectrometry (NIRS), which has the advantage of being non-destructive. The comparison with the Kjeldahl reference is done by a rather special chemometric treatment, partial least-squares (PLS) regression [6]. It is not exactly a validation but rather a calibration that allows to estimate a nitrogen content by referring to a database obtained by applying the Kjeldahl reference method to a set of samples close to the one being measured. Using this approach, the eventual bias of the Kjeldahl method is not detected and hence not corrected for.

References

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Conclusions

The role and interest of the measurements carried out and the results obtained, year after year, in the analytical laboratories are not denied. Both continuous improvement of analytical and bioanalytical methods and the importance of regulatory authorities to protect the population support this dynamic. Although it is difficult to put an exact figure on the number of results produced daily by the various laboratories, it is clear that it has been rapidly growing for several decades.

In the first chapter of this book, we have tried to explain the reasons for this growth, which are sometimes associated with a strong political and societal demand, but the response to this demand must be consistent.

The recent public health crisis associated with the SARS-CoV-2 virus (COVID-19) illustrates a fundamental requirement for the analytical sciences, namely the use of validated protocols and methods. For the first time in history, the administrative bodies in charge of public health have massively called upon laboratories to perform daily measurement to manage the pandemic. The success of this new strategy is open to question and should be evaluated.

The urgency of official demand has led analysts to a rush to use new methods with the risk of neglecting what had made their reputation since the 1990s, namely the standardization of procedures and inter-comparison of results and laboratories which are efficient in reducing measurement dispersion and discrepancies. During the first year of the crisis, numerous methods were developed and used without concern for harmonization. Whether to detect the virus and its variants, for example, by using reverse transcriptase polymerase chain reaction (RT-PCR) with several dozen probes that were proposed by various companies, or to quantify and/or detect the multiple antibodies produced by the virus, with the marketing of several dozen techniques and technologies in kit form.

The wealth of data produced has also raised many questions. For example, because of the lack of standardization, it was difficult after one year to use the first measurements obtained to understand the initial dynamics of the pandemic. As a result, by the end of 2020, several groups were already highlighting the risks resulting from the high frequency of false positives, but also of false negatives. Some even denounced part of the discourse about reinfections, which they explained precisely by measurement errors [1].